SPACE AND EARTH SCIENCE FOCUS AREA

(4) Space Science Core Courses + (2) Earth Science, (2) Heliophysics & Planetary Science, (2) Astrophysics Electives

CORE COURSES

Select 4 of the following, including 675.613, followed by 615.744

Required

- **675.613** Bold Science — Every Term • Asynchronous
- **675.744** Physics of Space Systems I — SU / FA • Asynchronous
- **615.641** Mathematical Methods for Physics & Engineering — Every Term • Asynchronous
- **615.642** Electromagnetics — FA / SP • Asynchronous
- **615.653** Classical Mechanics — Every Term • Asynchronous
- **615.654** Quantum Mechanics — FA / SP • Asynchronous
- **615.665** Modern Physics — SP / SU • Asynchronous
- **615.671** Principles of Optics — Every Term • Asynchronous
- **615.651** Statistical Mechanics and Thermodynamics — Every Term • Asynchronous
- **615.680** Materials Science — FA / SP • Asynchronous

EARTH SCIENCE ELECTIVES

Select 2 of the following

- **615.666** Earth Systems & AI — In Development • Async.
- **615.761** Introduction to Oceanography — SP • Asynchronous
- **615.769** Physics of Remote Sensing — SP • Asynchronous
- **615.775** Physics of Climate — SP • Asynchronous

HELIOPHYSICS AND PLANETARY SCIENCE ELECTIVES

Select 2 of the following

- **615.661** Introduction to Planetary Science — SP / SU • Asynchronous
- **615.663** Physics of Lunar Exploration — In Development • Async.
- **615.664** Introduction to Exoplanets — In Development • Async.
- **615.745** Physics of Space Systems II — SP • Asynchronous
- **615.755** Space Physics — FA / SP • Asynchronous

ASTROPHYSICS ELECTIVES

Select 2 of the following

- **615.662** Introduction to Astrophysics — FA • Asynchronous
- **615.748** Introduction to Relativity — SP • Asynchronous
- **615.772** Cosmology — SU • Asynchronous

HARDWARE ELECTIVES

OPTIONAL: Select up to 2 of the following to substitute other electives

- **615.747** Sensors and Sensor Systems — FA / SP • Asynchronous
- **615.763** Infrared Sensors: Physics and Applications — SP • Asynchronous

Test your degree path with the Applied Physics Course Mapping Tool